



Project Report on

Morse Code based SoS System using ASK Signalling Scheme

Submitted by

P.Lasya	23881A04J1
P.Karthikeyan Reddy	23881A04J2
P.Pranitha	23881A04J3

Department of Electronics and Communication Engineering

Vardhaman College of Engineering, Hyderabad

Contents

1	Introduction	3
2	Literature Review	3
3	Objectives	3
4	Methodology	4
4.1	Block Diagram	4
4.2	Circuit Diagram and Components	4
4.3	Working Principle	5
4.4	Software Implementation	6
5	Results and Discussion	6
6	Mapping POs and SDGs	7
6.1	Program Outcomes (POs) Mapping	7
6.2	Sustainable Development Goals (SDGs) Mapping	8
7	Conclusion	8
	Bibliography	9

Abstract

This project demonstrates a simple wireless communication system using Arduino and 433 MHz RF modules for transmitting and receiving emergency signals. The system is designed to send the Morse code for “SOS” — a universally recognized distress signal — from a transmitter Arduino to a receiver Arduino. The transmitter encodes the message “SOS” and sends it wirelessly using the RadioHead ASK library. On the receiver side, the Arduino decodes the incoming signal and controls an LED to blink in the Morse code pattern corresponding to SOS (· · · — — — · · ·). This provides a clear and recognizable visual indicator of the emergency message. The project highlights the use of low-cost RF communication, embedded C programming, and Morse code signaling for emergency alert applications. Such a system can be extended for disaster communication, rescue operations, and low-power wireless signaling in areas with limited infrastructure. The project demonstrates the integration of microcontrollers, RF communication modules, and Morse code encoding for the purpose of emergency communication. Unlike conventional continuous LED blinkers, the SOS pattern provides a meaningful signal recognizable by both humans and automated detection systems. The design emphasizes simplicity, low power consumption, and ease of replication, making it suitable for educational purposes as well as practical use in rescue missions, disaster-prone areas, and low-infrastructure environments. Through this implementation, the project also illustrates key concepts of wireless data transmission, digital signal processing at the microcontroller level, and the application of historic Morse code in modern embedded systems. The system can be extended to support other types of distress messages, incorporate buzzer or sound-based signaling, and integrate with larger Internet of Things (IoT) networks for enhanced safety and emergency communication applications.

Keywords: Arduino, 433 MHz RF module, RadioHead ASK library, Wireless communication, Morse code (SOS), Embedded systems, Emergency signaling, Distress communication, LED indicator, Low-power transmission, ASK modulation.

1 Introduction

Communication plays a vital role in ensuring safety and coordination during emergency situations. In scenarios where conventional communication systems such as mobile networks or the internet may fail, alternative methods of signaling become necessary. Morse code, one of the earliest forms of coded communication, is still recognized worldwide for transmitting simple but critical messages, such as “SOS,” which universally indicates distress. This project aims to design and implement a simple yet effective wireless emergency signaling system using Arduino microcontrollers and 433 MHz RF modules. The transmitter unit is programmed to repeatedly send the message “SOS” in Morse code format using the RadioHead ASK communication library. The receiver unit decodes the transmitted data and controls an LED, which blinks according to the SOS pattern (· · · — — — · · ·). This approach creates a low-cost, low-power, and portable system capable of delivering a clear visual distress signal.

2 Literature Review

1. Arduino Education — “Morse Code Project” (2025). This official Arduino education page provides a practical, beginner-oriented step-by-step implementation of a basic Morse encoder/decoder using Arduino boards and standard peripherals (LEDs, buttons, buzzers). Its main value is pedagogical: clear wiring diagrams, sample code, and learning objectives. It’s strong for teaching circuit setup and basic timing logic, but it typically avoids rigorous evaluation (accuracy metrics, noise robustness) and does not address scaling to wireless or noisy real-world channels.

2. Silva et al., “Morse Code Translator Using the Arduino Platform” (SAI Conf., 2016). Silva and colleagues present an implemented translator running on Arduino that converts user input to Morse and vice versa. The paper documents design choices (timing thresholds, lookup tables) and demonstrates a working prototype. Compared to [1], it is slightly more formal and provides implementation details useful for replication. Limitations include a lack of systematic testing against variable input speeds, limited discussion of debouncing or asynchronous input, and little treatment of automated detection (i.e., from audio or RF signals).

3. Akare et al., “Morse Code Detector Using Machine Learning” (IJARCCE, 2024). This study applies ML methods to detect Morse from sampled signals. The authors extract time-domain features and use classifiers to distinguish dots/dashes and inter-symbol gaps, reporting improved detection under noisy conditions compared to simple thresholding. This work is valuable because it addresses robustness and automates parameter tuning. Weaknesses to note: the paper may rely on relatively small or synthetic datasets, and details on computational cost (run-time, memory) for deployment on constrained microcontrollers are often limited or absent — an important consideration for Arduino-class hardware.

4. A. Javed, Building Arduino Projects for the Internet of Things (Apress, 2016). Javed’s text is a practical guide on integrating Arduino projects with networks, cloud services, and higher-level software. While not Morse-specific, it provides design patterns, libraries, and architectures for moving sensor or encoder data into IoT stacks (MQTT, REST APIs). This is critical if the project aims to transmit Morse translations to dashboards or a monitoring backend.

3 Objectives

- To design a wireless communication system using Arduino and 433 MHz RF modules.
- To transmit the SOS distress signal in Morse code format..
- To receive and decode the transmitted signal using Arduino.

- To provide a clear visual indication of SOS through LED blinking.
- To explore the application of Morse code in modern embedded systems for safety and rescue operations

4 Methodology

4.1 Block Diagram

The system as shown in figure 1 consists of:

- **Arduino Uno:** Microcontroller for processing and controlling the display.
- **433Mhz:** Receives the data
- **433Mhz Transmitter:** Transmits The Data
- **Antenna:**Data Transmitter

4.2 Circuit Diagram and Components

4.2.1 Components Used

- Arduino Uno
- 433Mhz Rf Receiver
- 433Mhz Rf Transmitter
- Helical Antenna

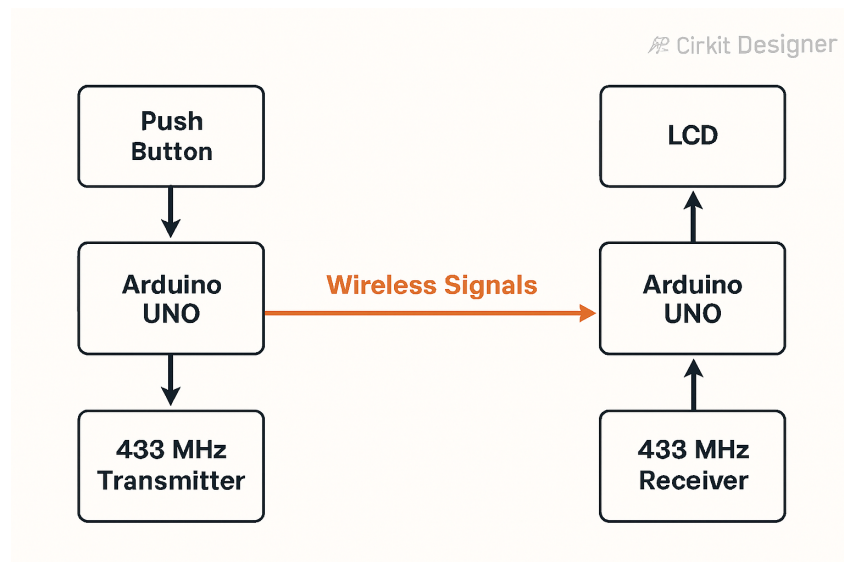


Figure 1: Block diagram : SOS Emergency System

The hardware of this project consists of two main sections as shown in figure 2: the transmitter and the receiver. At the core of both sections is the Arduino Uno, which acts as the main controller. The transmitter unit includes a 433 MHz RF transmitter module connected to the Arduino. This module is responsible for wirelessly sending the encoded SOS message using Amplitude Shift Keying (ASK) modulation. The receiver unit uses a 433 MHz RF receiver

module that captures the transmitted signal and forwards it to the Arduino for decoding. An LED is connected to the receiver Arduino to visually display the SOS Morse code pattern , providing a clear emergency indication. Both transmitter and receiver circuits are powered either through the Arduino's USB connection or an external 5V power supply, with the RF modules drawing power directly from the Arduino's 5V output. The RF module broadcasts this signal wirelessly, and the receiver module, tuned to the same frequency, captures it. The receiver Arduino reads the incoming signal, decodes the sequence, and triggers an output like blinking an LED or sounding a buzzer whenever the SOS pattern is detected. A capacitor across VCC and GND of the RF modules can help stabilize the power supply and prevent signal errors.

4.3 Working Principle

The SOS Transmission Using ASK Modulation system works on the principle of varying the amplitude of a carrier wave according to a digital SOS signal. When an emergency occurs, the SOS signal is triggered either by pressing a push button or automatically through a sensor. This signal is processed by a microcontroller, which generates a digital pattern representing the SOS code (in Morse format "... — ..."). The digital data is then given to an ASK (Amplitude Shift Keying) transmitter module, where the amplitude of a high-frequency carrier signal is modulated—meaning the carrier is transmitted for logic '1' and turned off (or reduced) for logic '0'. This modulated RF signal is transmitted wirelessly through the air using an antenna. On the receiver side, an ASK receiver tuned to the same frequency receives the signal and demodulates it by detecting the amplitude changes, thereby recovering the original digital SOS message. The received signal is then decoded by another microcontroller, which verifies the SOS pattern and activates an alert such as a buzzer, LED, or display message indicating distress. Thus, the system enables reliable wireless transmission of emergency SOS signals using the amplitude variation principle of ASK modulation.

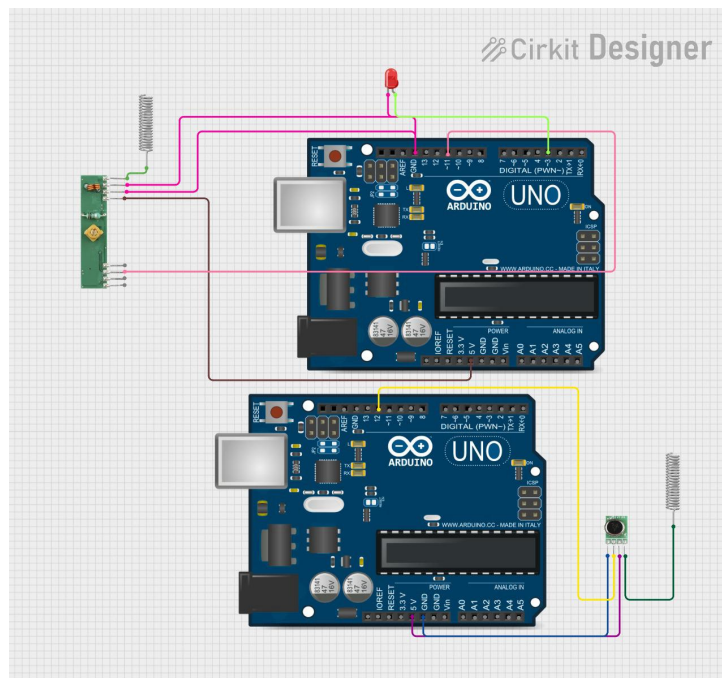


Figure 2: Circuit diagram : SOS Emergency System

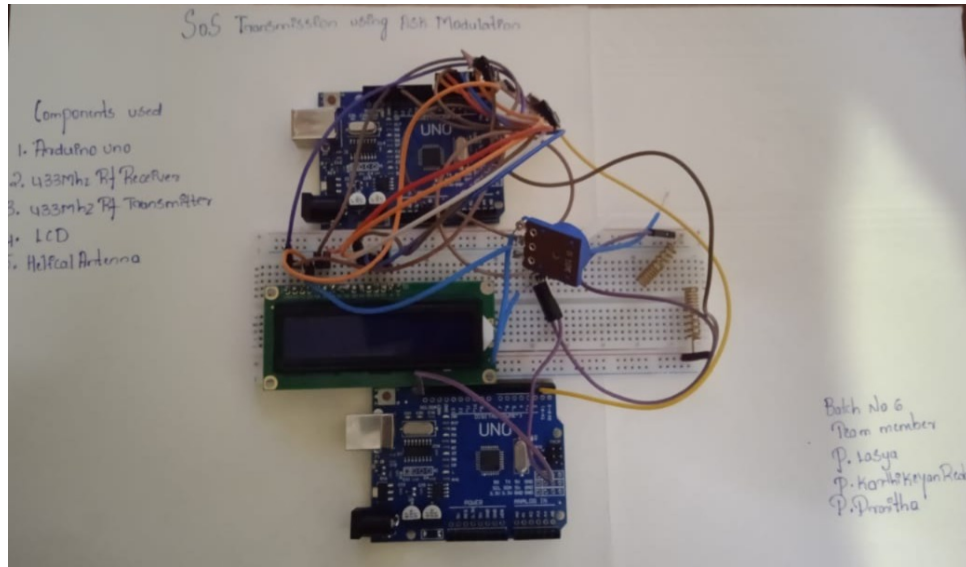


Figure 3: Prototype

4.4 Software Implementation

- **Arduino IDE:** Used to program the Arduino microcontroller.

5 Results and Discussion

The system successfully transmitted the SOS message using the 433 MHz RF transmitter and received it with the RF receiver as shown in figure 3. The receiver Arduino accurately decoded the message and blinked the LED in the correct Morse code pattern. Tests showed reliable communication within 10–15 meters indoors and longer ranges outdoors. The project proved to be low-cost and effective, though limited by range and interference. Overall, it demonstrates the feasibility of using simple RF communication for emergency signaling.

During practical testing, the communication range was observed to be around 10–15 meters indoors, where walls and interference reduced the effective distance. In open outdoor conditions, the range increased to approximately 50–100 meters, demonstrating the potential of the RF modules in less obstructed environments. The accuracy of the Morse code timing was maintained across repeated trials, showing that the system was consistent and reliable for its intended function.

6 Mapping POs and SDGs

6.1 Program Outcomes (POs) Mapping

Program Outcomes (POs) with relevance to our project “Quadrature Amplitude Modulation (QAM) in LTE and 5G Wireless Systems” as shown in Table 1.

Table 1: Mapping of Program Outcomes (POs) to Project Relevance

PO No.	Program Outcome	Relevance
PO1	Engineering Knowledge	Applied fundamentals of electronics, embedded systems, and communication engineering to design a Morse code transmitter and receiver using Arduino and RF modules.
PO6	Engineer and the World	Demonstrated a low-cost wireless solution applicable in real-life emergency communication scenarios.
PO8	Ethic	Ensures responsible design for safety-critical applications.
PO9	Individual and team work	Collaborates in hardware/software integration and testing.
PO11	Lifelong learning	Encourages further study in embedded systems, IoT, and communication technologies.
PSO2	Embedded system design	Designs and implements embedded communication solutions using ASK and Morse code.

6.2 Sustainable Development Goals (SDGs) Mapping

Sustainable Development Goals (SDGs) with their relevance to your project “Quadrature Amplitude Modulation (QAM) in LTE and 5G Wireless Systems” in Table 2.

Table 2: Mapping of Sustainable Development Goals (SDGs) to Project Relevance

SDG No.	Goal	Relevance
SDG 3	Good Health and Well-Being	Sends emergency SOS signals to save lives in accidents or remote areas.
SDG 9	Industry, Innovation, and Infrastructure	Promotes smart infrastructure in educational institutions.
SDG 11	Sustainable Cities and Communities	Reduces paper wastage by replacing traditional notices.
SDG 13	Climate Action	Enables reliable SOS signaling during climate-related disasters.

7 Conclusion

The project demonstrates the integration of engineering knowledge with ethical and sustainable practices, aligning with relevant Program Outcomes (POs) and Sustainable Development Goals (SDGs). The designed wireless SOS communication system showcases practical application, innovation, and responsible engineering, contributing to public safety and smart infrastructure development.

This project successfully demonstrated a simple wireless emergency signaling system using Arduino and 433 MHz RF modules. The transmitter was able to send the message “SOS” continuously, and the receiver accurately decoded it to blink an LED in the Morse code pattern ($\cdot \cdot \cdot - - - \cdot \cdot \cdot$). The results confirmed that the system is low-cost, power-efficient, and effective for short-range communication.

Bibliography

- [1] Arduino Education, “Morse Code Project,” *Arduino Official Education Page*, Apr. 27, 2025. Available: <https://www.arduino.cc/education/morse-code-project/>. [Accessed: Sep. 09, 2025].
- [2] S. S. Silva, A. Valente, S. Soares, M. J. C. S. Reis, J. Paiva, and P. Bartolomeu, “Morse Code Translator Using the Arduino Platform,” in *Proc. SAI Computing Conference*, London, UK, Jul. 2016, pp. 1–6.
- [3] U. P. Akare, K. Bhure, A. Sonkusre, A. Ganorkar, M. Barapatre, and P. Roychowdhury, “Morse Code Detector Using Machine Learning,” *Int. J. Adv. Res. Comput. Commun. Eng.*, vol. 13, no. 4, Apr. 2024. doi:10.17148/IJARCCE.2024.13414.
- [4] A. Javed, *Building Arduino Projects for the Internet of Things*, 1st ed., Shelter Island, NY: Apress, 2016.